

EEET202

INES & TRANSFORMERS

MODULE:3

Performance Characteristics of DC Motors

of operation of DC motors and select appropriate motor types for

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Syllabus

generation of torque, torque equation, performance characteristics – starters – 3 point and 4 point starters (principle only).

tors - field control, armature control. Braking of dc motors. Power and efficiency. Testing of dc motors - Swinburne's test, Hopkinson's test. DC motor applications – numerical problems.

Performance Characteristics

Characteristics (ω vs I_a): It shows the relation between armature current for constant applied voltage.

Characteristics (T_a vs I_a): It shows the relation between in the armature with armature current for the voltage.

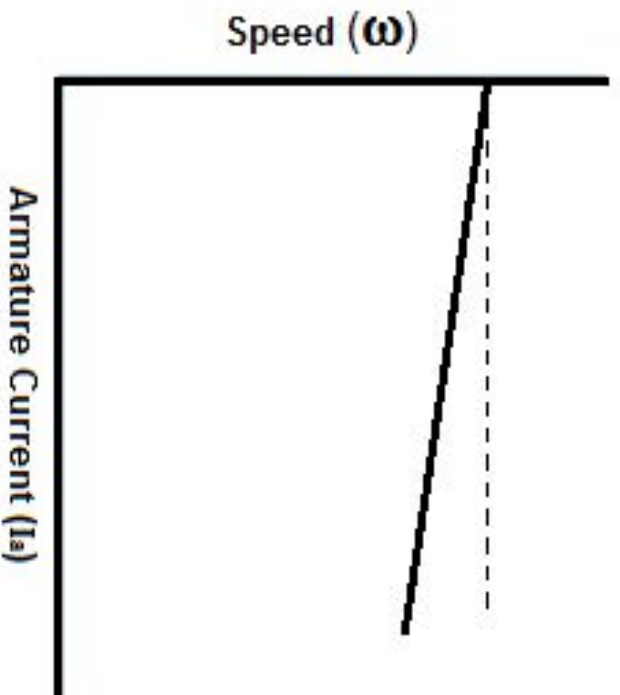
Characteristics (ω vs T_a): It shows the relation between torque developed by the motor for constant applied

I_a torque vs current characteristics together known as **mechanical characteristics** of dc motor and speed vs torque characteristics **operating characteristics** of dc motor.

Characteristics of Shunt Motor

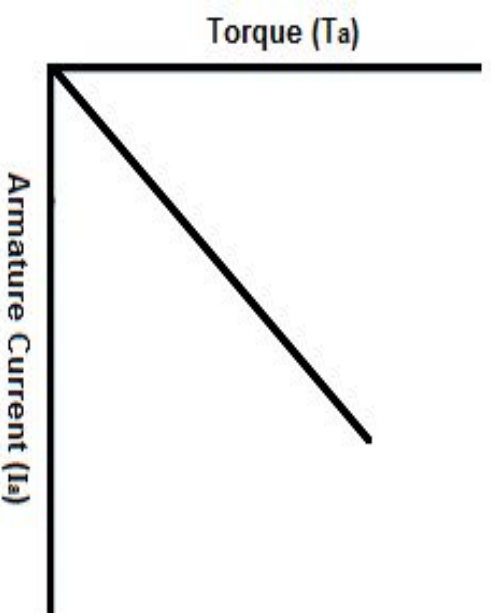
Characteristics (ω vs I_a)

$$E_b = \frac{Z\phi N}{60} \frac{P}{A} = K\phi\omega$$



Characteristics of Shunt Motor

Characteristics (T_a vs I_a)

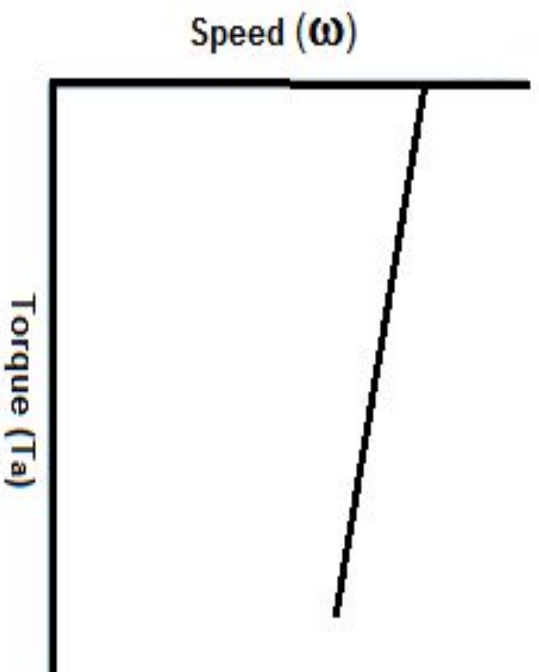


For a shunt motor, flux ϕ will be constant for a constant applied voltage. Therefore, the torque developed in the motor will be directly proportional to the armature current. Thus torque-current characteristics of shunt motor are shown in the figure.

Characteristics of Shunt Motor

Characteristics (ω vs T_a)

$$E_b = \frac{V - I_a R_a}{K\phi}$$



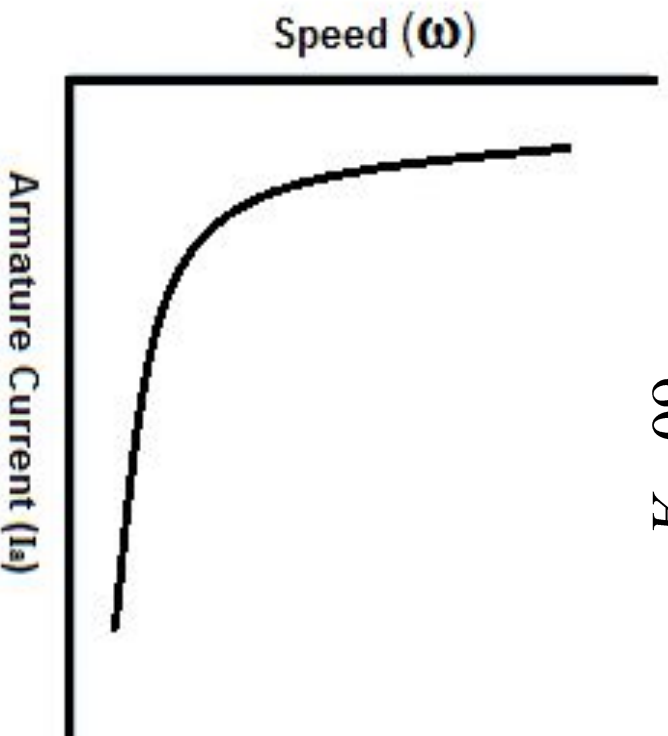
$$T_a = \frac{E_b}{K\phi}$$

As the load torque increases, the armature current I_a will increase, and the speed of the motor will decrease slightly. This is because the speed of a shunt motor is not perfectly constant but decreases slightly as the load increases. The characteristics of a shunt motor will be as shown in the figure.

Characteristics of Series Motor

Characteristics (ω vs I_a)

$$E_b = \frac{Z\phi N}{60} \frac{P}{A} = K\phi\omega$$

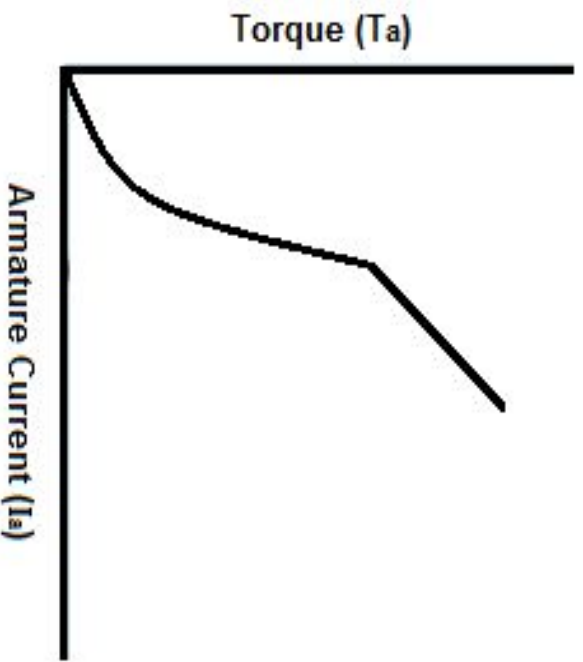


directly proportional to

$$\omega = \frac{V}{K_1 I_a} - \frac{R_a + R_{se}}{K_1}$$

Characteristics of Series Motor

Characteristics (T_a vs I_a)



$\phi \propto I_a$ will be proportional to armature current. Therefore torque of motor will be directly proportional to armature current.

For some large values of armature currents, flux will become unchanged because of the saturation of the magnetic core. Therefore, after the saturation of the core, torque will linearly vary with armature current.

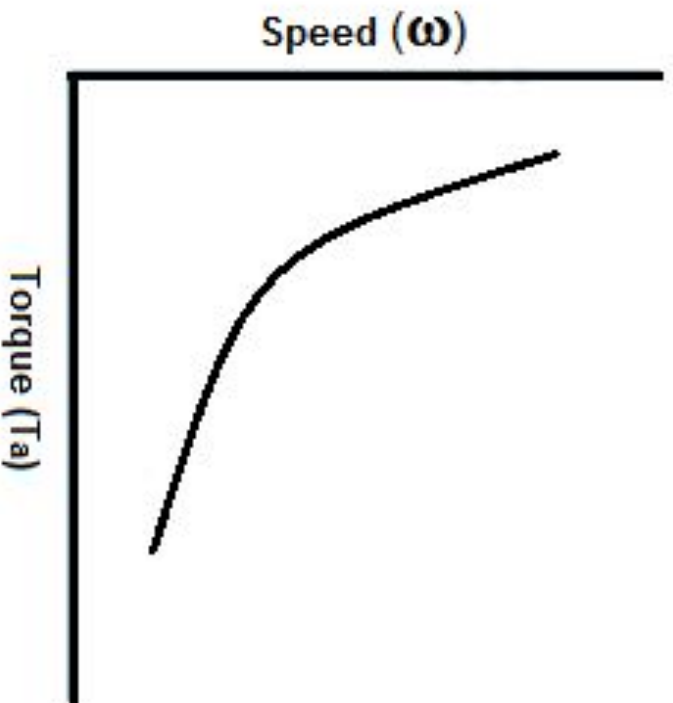
Characteristics of Series Motor

Characteristics (ω vs T_a)

$$\frac{-I_a(R_a + R_{se})}{K_1 I_a}$$

R_a

$\frac{R_a}{K_1}$



Starting of DC Motor

$$= \frac{Z \phi N}{60} \frac{P}{A} = K \phi \omega$$

DC motor,

so

Resistance is very low, armature current will be very large.
Voltage drop across armature windings and dip in supply voltage.

the use of starters.

Large resistance in the armature path, so that armature current is limited to a safe range during starting.

Starting of DC Motor

starters are:

arter

arter